

2.1 Computational Thinking

کمپیوٹیشنل تھنکنگ (Computational Thinking) مسائل حل کرنے کا ایک منظم طریقہ ہے۔

CS XI

Chapter 2: Computational Thinking & Algorithm

Lecture 01 of 13

2.1 Computational Thinking

LEARN IT

Computational Thinking (CT) is a systematic approach to solving problems. "Systematic" means you follow a fixed, organised method instead of guessing. CT involves four activities:

- **Decomposition** — breaking a big problem down into smaller parts.
- **Pattern recognition** — identifying things that repeat or follow a rule.
- **Abstraction** — hiding (ignoring) details that do not matter.
- **Algorithms** — developing clear step-by-step instructions to solve the problem.

CT is not "thinking like a computer". It is thinking in a way that lets *you* describe a solution so clearly that a computer (or another person) could carry it out.

Problem → Decomposition → Pattern recognition → Abstraction → Algorithm → Program

You already use computational thinking in daily life. When you plan your exam preparation, you split the syllabus into chapters (decomposition), notice that every chapter needs the same routine of reading, making notes and practising (pattern recognition), ignore things that are not in the syllabus (abstraction), and make a day-by-day timetable (algorithm).

LEARN IT

Did you know? Pseudocode is a human-readable representation of an algorithm using plain English and structured programming keywords. It is the bridge between human logic and machine code. You will study it in detail in section 2.2.2.

REMEMBER IT

Remember the four parts of CT with the phrase "**Don't Pick Any Apple**": Decomposition, Pattern recognition, Abstraction, Algorithm.

COMMON MISTAKE

Students sometimes think computational thinking means "writing code". It does not. CT happens *before* coding. You can do all four steps with pen and paper, without a computer.

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کمپیوٹیشنل تھنکنگ (Computational Thinking) مسائل حل کرنے کا ایک منظم طریقہ ہے۔ اس کے چار حصے ہیں: مسئلے کو چھوٹے حصوں میں توڑنا (Decomposition)، دہرائی جانے والی باتوں کو پہچاننا (Pattern Recognition)، غیر ضروری تفصیلات کو چھپانا (Abstraction) اور مرحلہ وار حل لکھنا (Algorithm)۔ یہ سوچنے کا طریقہ ہے، کوڈ لکھنا نہیں۔

CHECK YOURSELF

1. Name the four parts of computational thinking.
2. Is computational thinking done before or after writing code?
3. What is pseudocode called a "bridge" between?

Answers: 1. Decomposition, pattern recognition, abstraction, algorithms. 2. Before coding (it is planning the solution). 3. Between human logic and machine code.

Next lecture: 2.2 Algorithm